

Gender Effects on Ambiguity Aversion

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Abstract (Written by: A. Lauber & I. Kerkman; Edited by: all)

Results of the Ellsberg paradox determined that people are aversive to ambiguous choices when given an opportunity to choose between a risky and ambiguous option. The present study aimed to determine if there is a point in which people will pick an ambiguous option, and if this differs between genders. The presented task involved a risky and ambiguous option, and every four times the participant clicked the risky option, the probability of winning decreased by ten percent. Results showed that women tend to switch to the ambiguous option slightly quicker than men and non-binary participants, though the difference is very small, and all participants tend to switch when the risky option is at 50%. As well, the results showed differences in preference for the ambiguous option across risk levels. Future studies looking into dopaminergic and NMDA receptor activity could shed further light onto the behavioral differences between genders that emerged in this study.

Introduction (Written by: A. Lauber; Edited by: all)

People make decisions every day. From deciding what to eat for breakfast to determining what the best place to go to college is, people are always weighing their options and deciding what is best for them in that circumstance. What makes people decide on one option over another? Is the statistically optimal choice always the best choice, or is it important to consider subjective factors such as your own personal beliefs and feelings? This question does not have one straightforward answer, and varies between people and situations. There are many factors that need to be considered when making decisions including what options are available, what are their benefits, risks, and what is ambiguous about a choice.

It is typically believed that the limiting factor to making an optimal decision is the uncertainty in the environment. Evidence for or against a certain choice may be coming from multiple sources, which needs to be integrated to form the optimal decision. Sometimes this evidence has equal weight, while other times, evidence may influence a decision more heavily than others.¹¹ All of these factors need to be considered when determining if a choice is optimal.

Decision theory is how people make optimal choices when encountering uncertainty. It accounts for all the available information a person has and allows for the best decision to be made by minimizing consequences.¹⁰ Based on the information collected, the mathematical optimal choice is considered to have the highest expected value. However, people do not always choose the mathematically best choice because of the subjective value they place on different things. When someone accounts for objective probabilities such as the risk of a certain choice and their subjective probabilities or personal beliefs, the optimal decision would have the greatest expected utility.² A person may choose an option with the highest expected utility even if it is not the ideal choice statistically and therefore doesn't have the highest expected value.

There are multiple types of uncertainty associated with the outcome of an event accounting for the risk and ambiguity of a choice. Aleatory uncertainty is a type of uncertainty that is objective and cannot change, no matter how much information you have about it. It is unpredictable and considered to be risky.⁹ For example, knowing there is a $\frac{1}{3}$ chance you will win a task would be aleatory uncertainty because the participant knows exactly what their chance of winning is. Epistemic uncertainty is a type of uncertainty that is subjective and is due to a lack of knowledge. It would be more predictable with more information, and is considered to be ambiguous.⁹ For example when you walk into a restaurant without a reservation, you will not know how much time it will take to be seated.

The Ellsberg paradox is a task that involves decision making, and involves both aleatory and epistemic uncertainty. Participants are asked to choose between two gambling options, one in which they know the probability of winning (risk), and the other they do not (ambiguous). The main finding with the Ellsberg paradox is that people are more likely to choose the risky option over the ambiguous one. This is called ambiguity aversion, and could be due to people trying to avoid situations where they are less informed than others.³

How does ambiguity aversion differ between people and between gender? Past research shows mixed results. Some studies find that women are more risk and ambiguity averse while men make unpredictable choices.¹ However, this can change based on context and how the outcome is structured. For example, when put in a group

or social setting, both men and women show more ambiguity aversion.⁶ It can also depend on whether the choice comes after losing a previous investment. After a loss, men are more likely to seek out risk compared to women.

The present study tests at what risk probability people will pick the ambiguous choice, and if that differs between genders. We hypothesize that as the probability of reward decreases with the risky option, participants will switch to the ambiguous option to potentially have a greater chance of reward. We also predict that men will display less ambiguity aversion and more risk-seeking behavior by switching to the ambiguous option more quickly than women. We also hypothesize that the average threshold at which participants switch overall will be around 50%.

Methods (Written by: C. Cassady; Edited by: all)

A total of 54 participants completed the study. There were 25 men, 26 women, and 3 non-binary participants. The participants were assigned to play a web based gambling game that involved two options. One option contained risky choices and the other contained ambiguous choices. The goal was to have the participants choose the ambiguous option instead of risk, which differentiates it from the Ellsberg Paradox task. Another goal was to see if there is a difference between gender and the behaviors during the game. The games recorded what the participants decided in every trial as well as tracked the gender of each participant.

One option involved more risk. The odds started out at 70%, a high chance of reward, and after a participant chose it, the chance began to lower. Every 4 times the participant chose the risky option, the reward probability would drop by 10%. The game recorded which option the participant chose, and when they switched to the ambiguous option. Participants were informed of these probabilities and saw the change in chance drop as they continued. The second option in the game involved ambiguity, an unknown risk. The odds were averaged out at 60%. Participants did not know these odds. This remained constant and did not change over time. So, the weight of evidence was manipulated in the game. Having this greater weight of evidence provided clear information as to how and when participants are willing to sacrifice.⁸ Participants completed a minimum of 30 trials. In total, there were 1,611 trials. Males completed 722

trials, females completed 797 trials, and non-binary participants completed 92 trials. Using repeated measures provides high internal reliability for assessment. This is important considering the number of participants.

The point at which participants change from choosing the known risk as it rapidly falls to the unknown risk was measured and recorded, this was the dependent variable. The statistics of the men's behavior and women's behavior were compared, with the data being anonymous. A statistical analysis was run through a two-way ANOVA test.

Results (Written by: C. Cassady; Edited by: all)

The effect size of participants, risk level, and the interaction between gender and risk is measured in Table 1 by Partial Eta Squared and Cohen's f . The partial eta squared represents the percentage of variance in the switch point. The risk level is the most significant, showing that around 12.7% of participants choosing the ambiguous option was due to the change in risk level. The effect size, according to Cohen's standards, of gender is small. This indicates that participants generally have similar decision making thresholds, and there is only a slight difference. When analyzing risk level, there is a large effect size meaning that controlling the risk levels heavily impacts participant behavior. Lastly, the interaction between the two showed a small effect size. In Table 2, the p -values are measured, which indicate the significance. The data shows that gender and risk level have a high p -value, both being <0.001 . The interaction between the two variables had a significant p -value of <0.044 . Overall, there is significance within the variables and interaction.

A Two-Way ANOVA test was run to analyze the interaction between gender, the known risk level, and the ambiguous choice rate. The evidence does not support the hypothesis that men have a more sensitive decision making threshold. The results show that females switch to the ambiguous option slightly quicker than men and non-binary participants. The Two-Way ANOVA test, Figure 1, also represents that participants overall were likely to switch to the ambiguous option at a risk level of around 50%. This is also seen in Figure 2, which represents the mean risk level when choosing the ambiguous option. It is clear that the majority of participants switched to the ambiguous option at around 50%. The difference between genders and their risk taking is relatively

small, however, there is a slight difference. Like stated above, women switched to the ambiguous option slightly quicker. Along with this, as seen in Figure 1, women are also more ambiguity tolerant. This means that the female participants stick with the unknown risk longer than other participants. These results suggest that gender does have a slight effect on the Ellsberg Paradox.

Table 1

Source of Variation	Partial Eta Squared (η^2)	Cohen's f	Interpretation (Cohen's Standards)
Gender	0.0095	0.098	Small
Risk Level	0.1269	0.381	Large
Interaction (Gender x Risk)	0.0150	0.123	Small

Note. Effect size of gender, risk level, and gender x risk interaction.

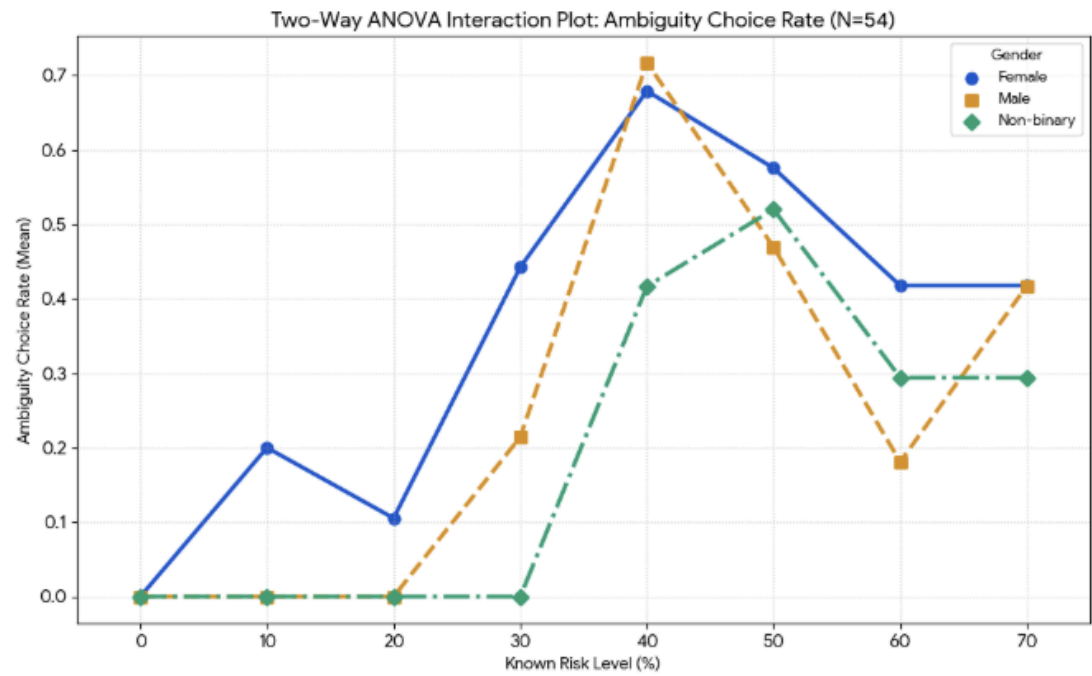
Table 2

1

Source of Variation	Sum of Squares	df	F-statistic	p-value	Significance
Gender	3.25	2.0	7.59	< 0.001	*** (High)
Risk Level	49.38	7.0	32.94	< 0.001	*** (High)
Gender x Risk	5.18	14.0	1.73	<0.044	* (Significant)
Residual	339.84	1587.0	-	-	-

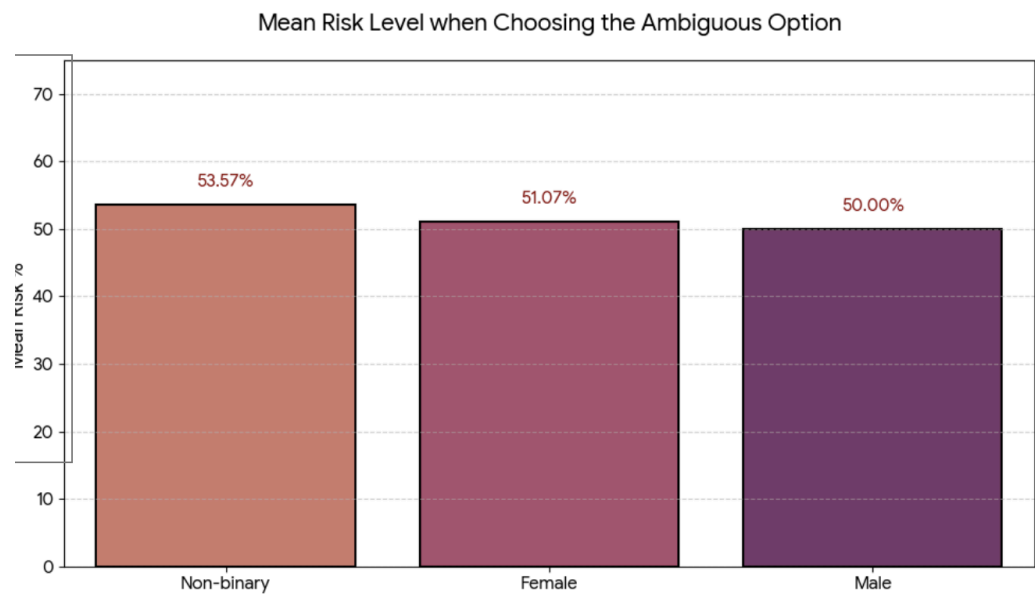
Note. Significance of p-values.

Figure 1



Note. Two-Way ANOVA interaction plot evaluating the ambiguous choice rate.

Figure 2



Note. Mean risk level of when participants switch to the ambiguous option.

Discussion (Written by: I. Kerkman; Edited by: all)

This study explored the effects of gender on ambiguity aversion and risk taking capability. Our primary hypothesis was fulfilled, as participants choose the ambiguous option much more often in the 40-50% range of known risk compared to any other known risk range. We also hypothesized the correct threshold of 50% known risk to switch the primary button from the known option to the ambiguous option. However, our secondary hypothesis that males would show less ambiguity aversion compared to females predicted a much simpler occurrence than what actually occurred. By comparing the likelihood of a participant to click the ambiguous risk button at different levels of known risk we can see two different behaviors towards risk taking by each gender. This comparison adds a layer of nuance onto the secondary hypothesis.

Further exploring the relationship between male, female, and non-binary participants, one can see that the difference lies not only in preference for the ambiguous option, but also in the exploration and exploitation of the buttons as well. Exploratory behaviors entail pressing each button to test to see which has the better odds, while exploitative behaviors entail pressing the button that you think will give you the best odds to score more in the study.⁵ Female participants tended to have exploratory behaviors in relation to the experiment, explaining the fact that their use of the ambiguous button is more spread out across the known risk levels compared to male and non-binary participants. Males tended to have more exploitative behaviors in the experiment, as they initially started at 40% usage of the ambiguous option at the 70% known risk level, the usage then dips until the known risk reaches below 50%, causing them to move to the ambiguous options and stick with that option. That behavior signals exploitative behavior because they stick with the option they believe will give them the best chance of doubling their score. The trend for non-binary participants is similar to male participants, just with a less sharp peak at 50%, also signalling exploitative behaviors.

Some limitations of this study include the fact that the sample may not be representative of the broader population because most participants are college students. Another is that because participants had no stake in the experiment, they may not provide valid results, and took either the perceptually riskier choice because they

wouldn't lose anything or the perceptually safer choice because there is no reason to take a risk because they would not gain anything. Hence to improve the study, making the participants earn their compensatory money through risk taking may be a good way to increase their want to participate, as well as sampling a better distribution from the broader population. As well, our results for the non-binary group may not be accurate so sampling more to gain a better representation of all genders would be helpful to gain new insights. As well, the function of the experiment could be improved if the game did not quit once the participants went below a certain amount of money so participants would be retained longer.

The emergence of exploitative and exploratory behaviors in this study provide a new pathway for researchers to explore. Perhaps a new study linking these behaviors, gender, and risk taking would allow for a more in depth analysis of the relationship rather than the relationship between the variables that this study found while testing different hypotheses. This implication of linking this set of behaviors together and risk taking would provide a new level of understanding as to certain groups' risk taking preferences over others.

Neuroscience Extensions (Written by: I. Kerkman; Edited by: all)

Past research had implicated dopaminergic activity in the encoding of both risk probability and uncertainty.⁷ The role of dopamine in uncertainty suggests that dopamine has a role in risk-taking signals,⁷ and by redoing this study while participants are doing a PET scan, we would be able to observe this possible relationship within the brain. It would also let us gather if dopamine's role varies by gender in a risk-taking task. This study provides an excellent opportunity to test this because the results from the study already conducted provide reason to believe that there will be significant differences in dopaminergic activity as there are already different patterns of behavior.

But however, the different patterns of behavior vary not in the mean known risk when the ambiguous button was pressed, but rather in the expression of behavior of pressing the ambiguous button. Females demonstrated exploratory behaviors, while males demonstrated exploitative behaviors. Because NMDA receptor-dependent activity of the brain's NA neurons can be used to see the balance of exploratory and

exploitative behaviors, ⁴ further testing using a PET scan would allow researchers to know more on whether males and females demonstrate different behavior practices while taking risks and what causes the differences in behavior in the brain. Our study would predict that there would be significantly different brain region activation between genders while going through this task.

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Contribution statement:

Code experimental program: Anna

Presentation: All

Abstract: Anna and Ian

Introduction: Anna

Methods: Charlie

Results: Charlie (report) and Ian (slides/statistics)

Neuroscience integration/limitations: Ian (report) and Charlie (slides)

Discussion: Ian

References: All